# Introduction to Neurological Assessment

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# Learning Outcomes

By the end of this lecture, students will be able to:

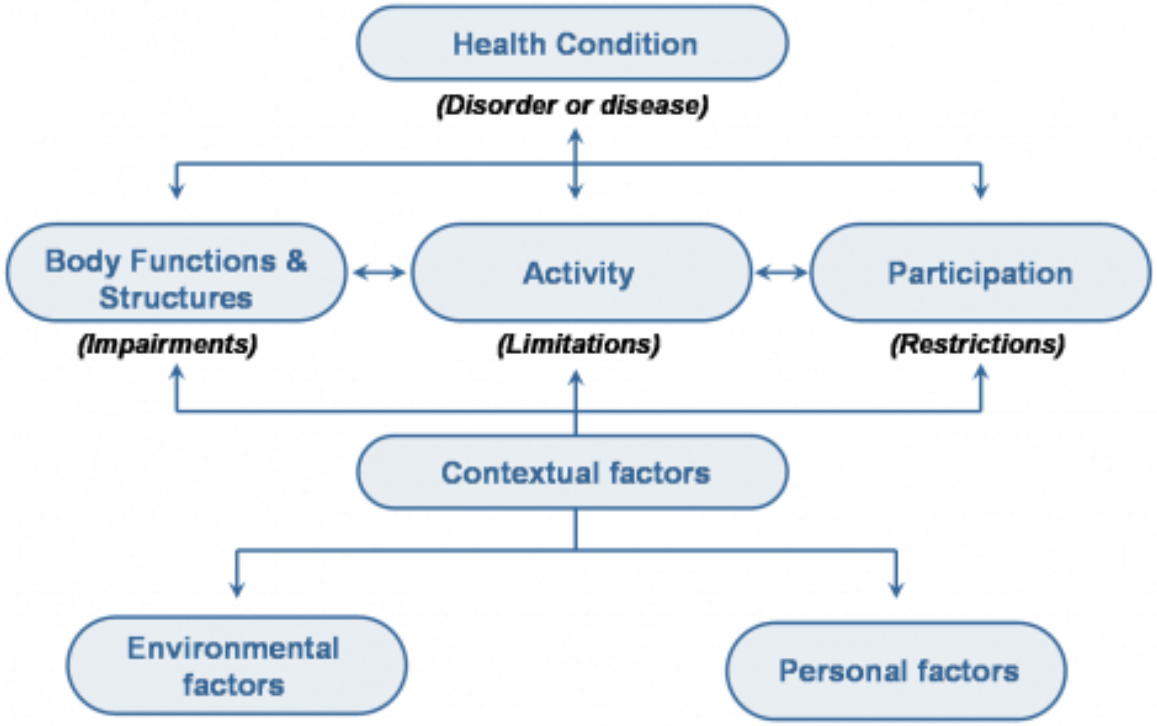
- Describe the components of a neurological assessment
- Discuss the ‘what’, ‘why’ and ‘how’ to conduct subjective and objective assessments in Neurological physiotherapy practice
- Explain how to utilise assessment findings in clinical reasoning

# Physiotherapy Neurological Assessment

- Context
  - First step in the rehabilitation process
  - Treatment efficacy is a good as the assessment on which it was based
  - Continuous process
  - Identify patient's problems, setting of patient centred goals, and personalised treatment plan

- Neurological problems
  - Multi-faceted encompassing **physical, psychological, cognitive** issues
  - Whole person consideration is necessary

# International Classification of Functioning, Disability and Health (ICF, 2001)



# Why Assess?

- **“assessments provide information on which clinical reasoning will be based and upon which decision making is reliant.”** (Jones, 2011)
- “Treatment efficacy is only as good as the assessment it is based on, and we only tend to treat what it occurs to us to assess.” (Keshner, 1991)
- Assessment is an ongoing & continuous process

# General aims of assessments in neurology

To establish:

- Baseline information (severity and ICF domains affected)
- Safety and independence
- Goal setting
- Patient's capacity to progress (to more complex actions, activities & discharge destination)
- Evidence of efficacy of treatment intervention

# Physiotherapy Neurological Assessment (components)

Subjective Assessment

Objective Assessment

# Subjective Assessment – what is it?

## What is it?

- Gathering information about the patient, and
- How their current condition affects them as a person

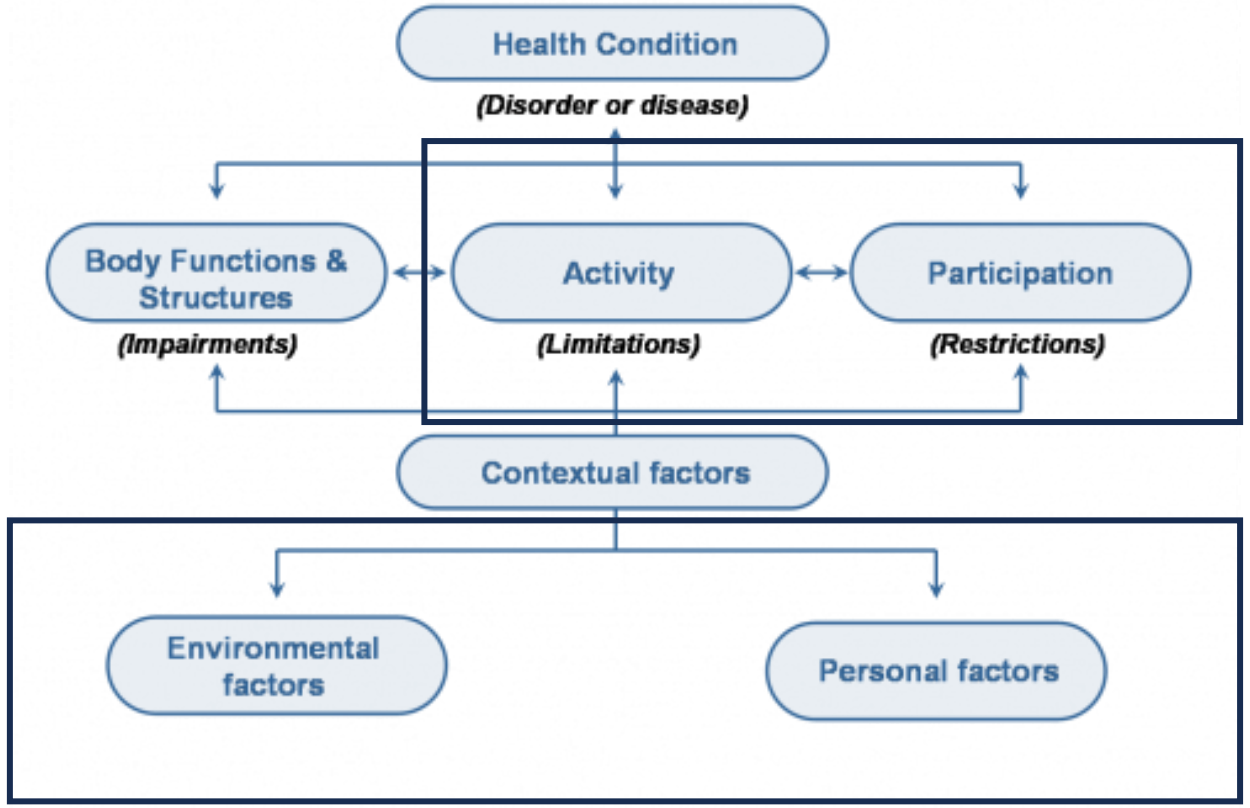
## Aim

- A detailed picture of how the present condition affects the patient from a holistic viewpoint (think ICF)

## Scope

- Personal to the patient – (patient, relative, carer or any other member of the MDT)

# International Classification of Functioning, Disability and Health (ICF, 2001)




**Sources:**

- MDT notes,
- GP letters
- Conversation with relative
- Patient interview

# Subjective assessment – the why?

- To identify the main problems as perceived by the patient (from patient's perspective)
- To assist the setting of short- and long-term goals (in collaboration with the patient)
- To assist in the development of a relevant treatment/management plan for the patient
- To build a rapport with the patient.

# Subjective Assessment – the how?

- Where do I start?
- Informed consent – legal, professional and ethical obligation.
  - Issues around consent?
- Data collection
  - Split over multiple sessions
- Main categories of a subjective assessment and why do I need to ask these questions?
- Where will I get this information from?
- What might I expect to see and hear from this patient
- What might be some difficulties and how will I get around these

# Contents of a subjective assessment

- **Personal Information**
- **Informed consent**
- **Present complaint**
- **History of present complaint**
  - Date of onset of the signs/symptoms
  - Progression of the condition
  - Investigation and results
  - Medical management, medical observations
  - Other management
  - Previous therapy
- **Past Medical History**
  - Co-morbidities and other non-related conditions
  - Special equipment, technology dependency
  - Previous surgery
- **Drug History**
  - Medications taken
- **Social History**
  - Personal Activities of Daily Living (PADL)
    - Washing, dressing, feeding/drinking, toileting
  - Domestic Activities of Daily Living (DADL)
    - Cooking, cleaning, shopping
- **Mobility**
  - General mobility: transfers?
  - Walking indoors, walking outdoors: how far, independence?
  - Stairs and steps: independence
- **Falls**
  - How many, what circumstances?
- **Environment**
  - Home, work and hobbies, accessibility
- **Psychosocial**
  - Personal factors, lifestyle, support, family situation

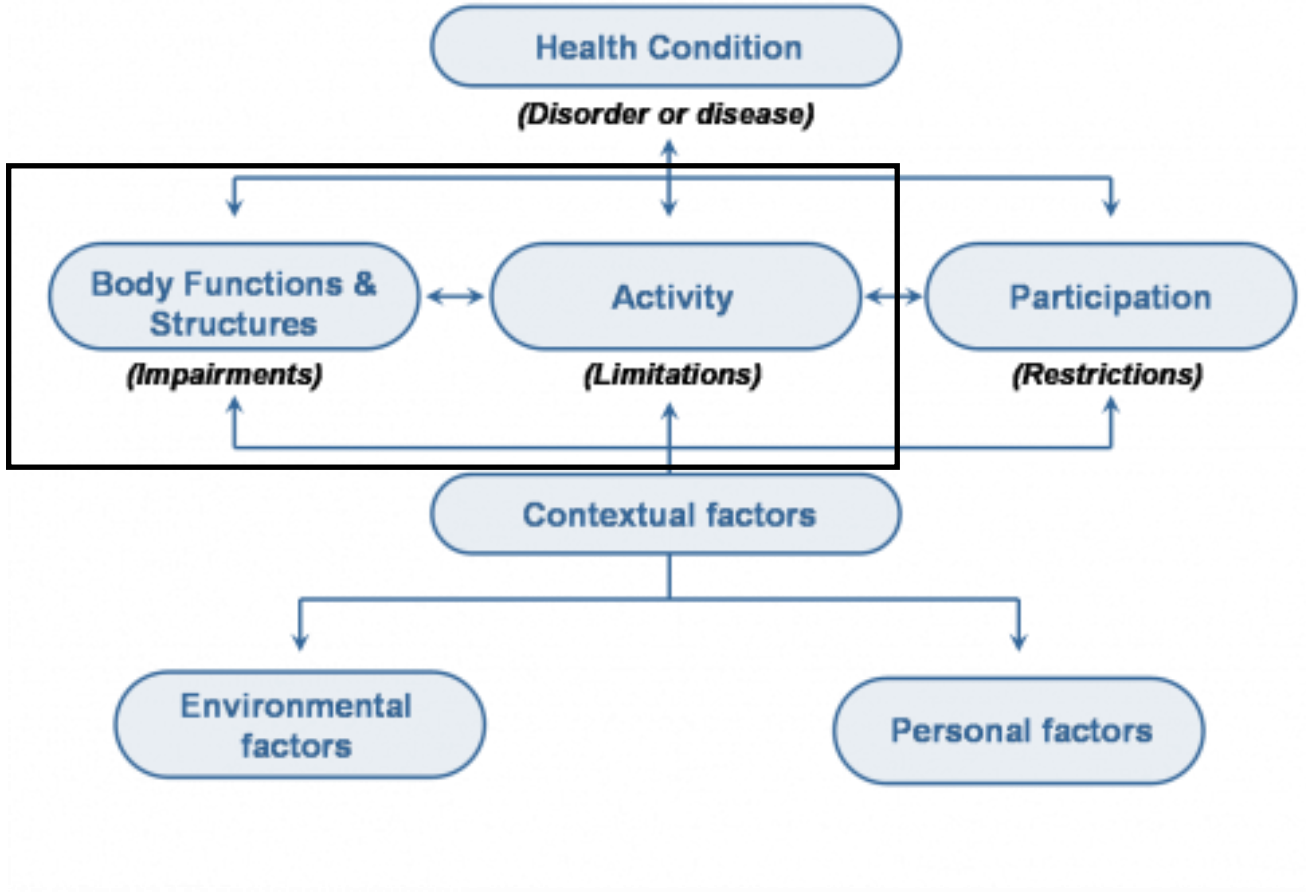
# Planning the objective assessment

- Analysis of subjective information obtained
- Determine:
  - clinical relevance of the provided information
  - Which functions are difficult for the patient and whether there are common factors influencing them
  - Break down functions to analyse the likely joint movement, muscle actions and motor control aspects
  - Reason the potential causes of the patient's difficulties based on the presenting pathology

# Objective Assessment – the what?

- Involves the physiotherapist gaining information related to the patient's movement disorder and functional status using measurable tools and movement analysis
- Assesses body structure and function, and activity domains of the ICF
- Generally
  - How patient moves (activities)
  - Reasons for the movement patterns or behaviours (body structure and function)
  - Uses observation and handling skills
  - Valid outcome measures

# International Classification of Functioning, Disability and Health (ICF, 2001)



# Objective Assessment – the why?

- Identify the movement problems and potential causes of those problems to appropriately focus treatment
- Providing a benchmark or baseline from which suitable short- and long-term goals can be agreed with the patient
- Provide a basis from which effectiveness of treatment can be evaluated.

# Objective Assessment – the how?

- Where do I start
- Risk Assessment
- Informed Consent
  - Touch and handling involved so a new informed consent required.
- The Objective assessment
  1. General observation
  2. Observation of how patient moves
  3. Investigation into why the patient moves as they do

- General impression of patient's condition, level of function based on the subjective
- Main categories in an objective assessment and what each tool tells me
- What assessment tools are a priority for the patient
- What may be the difficulties when assessing this type of patients and how to get around the issues
- Based on my risk assessment do I need assistance

# Summary of Objective Assessment

- **Risk assessment**
- **Informed consent**
- **General observation**
- **Functional Assessment** (Observation of how the patients moves during functional activities)
  - Gait
  - Functional activities
    - Mobility in bed
    - Transfers
    - Walking
    - Reaching to grasp
- **Investigations** of why the patient moves as they do

# General observations

- General Appearance
- Cognition and perception
- Vision
- Hearing
- Communication
- Behaviour
- Exercise tolerance

# Functional Assessment

- Observation of functional transfers
- Observation during functional tasks
  - dressing/undressing, reaching and grasp, writing
- Bed mobility
  - Rolling from side to side
- Sitting to lying
- Lying to sitting
- Sitting to standing
- Standing to sitting
- Walking indoors/outdoors
- Stairs
- Running

# What to assess in functional assessment

## Functional ability

- Success of the functional task
- What assistance they require to complete the task (assistance of 1 or 2) A01 or A02

## Quality of the movement

- Analyse the movement – looking for deviation from normal
- Focus on – any asymmetry, unco-ordinated movements, trick movements, gross patterning, involuntary movements, limitation in ROM, balance, lack of confidence, fatigue.

# Documenting your assessment

- Physiotherapists are legally responsible to keep an adequate record of the intervention to demonstrate to a third-party:
  - What
  - Why?
  - When of the assessment was completed.
- Forms of records
  - Written or electronic records
  - Video
  - Photos
  - Emails
- Impression
  - Useful to write a couple of notes giving an overall impression of the patient.

# Investigative tests

- Postural alignment
- Muscle tone
- Reflexes
- Sensation
- Dermatomes
- Trunk stability
- Co-ordination

- Vision
- Active/Passive Range of Movement
- Pain
- Strength/weakness
- Myotomes
- Balance
- Cognition and perception
- Falls

# Problem list

A problem is defined as anything that causes a concern to the patient

Patient's problems may be different to those of a PT's viewpoint.

Use the ICF to produce a problem list

**Impairments**

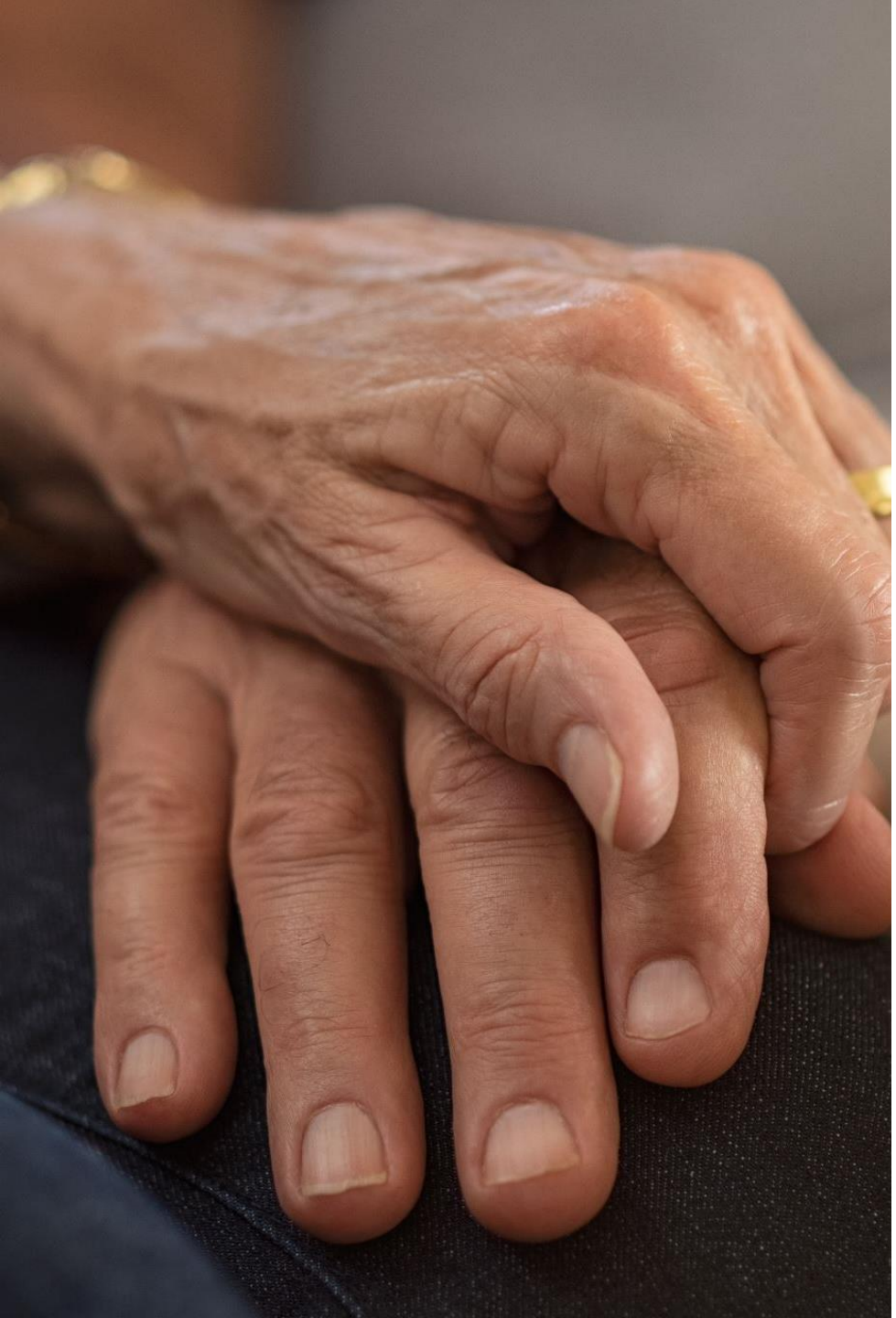
decreased range of motion, decreased muscle tone, low mood

**Activities limitations**

unable to sit-to-stand

**Participation restrictions:**

unable to carry out present employment as a primary school teacher.



# SMART patient centred-goals

- What is a goal?
- Discussed and agreed with the patient
- **SMART**
  - Specific – *focused and clear*
  - Measurable – *descriptive, quality and time element*
  - Achievable – *based on something the patient wants to do*
  - Realistic – *within the confines of available resources*
  - Timed – *by when?*

- Include short- and long-term goals
  - Short term: Mr A will be able to sit in an upright symmetrical posture, with stand-by assistance, for 30 seconds in 1 week
  - Long term: Mr A will be able to sit-to-stand in an arm-chair, weight bearing evenly through lower limbs, independently, in 3 weeks

# Further reading & Reference


- Jones, K. (2011) Neurological assessment: a clinician's guide. Edinburgh: Elsevier. Read Chapter 16 & 17
- Available at: http://www.vlebooks.com/vleweb/product/openreader?id=LondonMetr&isbn=9780702046605.